

[CO-1] Assessment Tool-2

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Course Code — MCA0401

Course — Deep Learning

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Assessment Tool-2

Introduction

Neural Network is a machine learning model that learns pattern from data and predict output

During Training - Model learns

During Testing - predicts result

Training MSE = 0.85

Testing MSE = 5.20

MSE (mean squared error)

$$MSE = \frac{1}{n} \sum (\text{Actual} - \text{predicted})^2$$

Lower MSE - better predicted

Higher MSE - Large predicted error

Given

Training MSE = 0.85

Testing MSE = 5.20

$$5.20 > 0.85$$

The model has high Testing error

∴ It is overfitting

⇒ That learns patterns from data and predicted the output

⇒ In that model learns from training and model tests and predict the results

Reasons for High-Testing Error

- ⇒ Over fitting
- ⇒ Small Training Data set
- ⇒ Complex Neural Network
- ⇒ Noise in Data
- ⇒ Lack of Regularization
- ⇒ Improper Data Splitting

Corrective measures

- ⇒ Increase Training Data
- ⇒ Use Regularization
- ⇒ Early stopping
- ⇒ Split the model
- ⇒ Remove Noise
- ⇒ Tune Hyperparameters

Training Error	Testing Error	Interpretation
High	High	underfitting
Low	High	overfitting
Low	Low	Good model

This model is overfitting

Conclusion

Since the testing error is much larger than the training error, the model is suffering from overfitting.